



PRIYABRATA DUTTA

Senior AI Engineer · Full Stack | GenAI / LLM / RAG · React / Next.js / Node.js / FastAPI / AWS

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer (6+ yrs, React/Next/Node/FastAPI/AWS) transitioning into GenAI application engineering — RAG, vector search, and LLM workflows shipped end to end. Technical Lead who turns architecture depth into product outcomes.

TECHNICAL SKILLS

AI & LLM Engineering: Generative AI application development, LLM integration (OpenAI API), RAG pipelines, LangChain, ChromaDB (vector DB), Prompt Engineering & Tooling, Agentic AI workflows, FastAPI for AI services

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, SSR / SSG / CSR, Responsive / Mobile-First Design

Backend: Node.js, Express, FastAPI, Python, REST API design, Authentication (JWT / OAuth), Payment Gateway Integration (Razorpay / Cashfree), Prisma ORM

Database: SQL, PostgreSQL, Redis (schema design, query optimization, indexing, connection pooling)

Cloud & DevOps: AWS Certified Solutions Architect – Associate, AWS (EC2, S3, CloudFront, Lambda), Docker, Kubernetes, CI/CD Pipelines, Git / GitHub

Leadership: Team Leadership, Agile / Scrum, Code Review & Mentoring, Project Management, Client Communication

PROFESSIONAL EXPERIENCE

Technical Lead / Full Stack Developer

Feb 2022 – Present

Appadd India (P) Ltd, Bangalore

- **Architected an LLM-powered reporting engine** (FastAPI, LangChain, ChromaDB) integrating GPT-4o via RAG: engineered custom prompt pipelines that automated complex data extraction and cut client report generation from 3 hours to ~10 minutes.
- **Engineered a multi-tenant Clinic Appointment SaaS platform (React, Node.js):** implemented complex state management to successfully eliminate concurrent double-bookings using optimistic locking and database-level unique constraints.
- **Built a comprehensive B2B e-commerce platform managing 5,000+ SKUs:** designed 35+ REST APIs with JWT/RBAC security and integrated idempotent Razorpay/Cashfree gateways to process ₹15 Lakh+ in monthly transaction volume.
- **Optimized full-stack performance across 8 client domains:** reduced catalog search latency to <40ms utilizing Redis and MySQL full-text search, and cut frontend load times by 74% (4.2s to 1.1s) via code-splitting and edge caching.
- **Led AWS cloud infrastructure and DevOps initiatives:** migrated legacy applications to AWS (EC2, S3, CloudFront, Lambda) to cut hosting costs by 40%, and established zero-downtime CI/CD (GitHub Actions, Docker) that accelerated deployments by 80%.
- **Directed technical discovery and scaled engineering operations:** translated business requirements into scalable roadmaps for cross-functional clients while mentoring a 5-engineer team through rigorous PR standards that reduced production bugs by 60%.

Freelance Full Stack Developer

2020 – 2022

GraphicxIT, Siliguri

- **Delivered 20+ custom web applications across 4 countries** (India, USA, UAE, UK) using React.js, PHP, and Laravel — 100% on-time delivery, zero post-launch critical bugs.
- **Sole engineer on a Laravel application for a government body** — 5K+ registered users, document management, secure member workflows; designed the full database schema and auth system independently.

- **Achieved Google PageSpeed 90+ across all client sites** through responsive design, image optimization, and performance tuning — contributing to a 30% average increase in organic traffic within 3 months.
- **Optimized SQL schemas reducing average query time by 65%** — one client's dashboard load dropped from 8s to 2.8s.
- **Automated 10+ client workflows with Zapier and custom REST integrations** — saving 15+ hours/month of manual data entry per client.

Technical Resource / Web Developer

2019 – 2020

Uttarbanga Sambad, Siliguri

Owned full-stack development and production infrastructure for one of North Bengal's largest news organizations — building and operating a high-traffic platform serving 100K+ readers, while leading a cross-functional engineering team.

- **Architected and built a high-traffic news portal (full-stack) serving 100K+ monthly readers** — engineered sub-second page loads that held under 3–5x traffic spikes during election and festival cycles.
- **Designed and maintained the production infrastructure end-to-end** — provisioned and tuned NGINX + MySQL, implemented WAF/SSL/DDoS hardening and automated backups, sustaining 99.5% uptime through peak election traffic.
- **Led a 20-member cross-functional team** across development and operations — introduced deployment and content workflows that cut publishing bottlenecks by 50%.
- **Drove a CMS migration and re-architecture** — scaled article throughput from 20 to 60+ articles/day (3x) with zero additional editorial headcount, through automation and pipeline optimization.
- **Built the streaming infrastructure for Facebook Live broadcasts reaching 10K+ concurrent viewers** — handled encoding, delivery pipeline, and failover.
- **Engineered an ad-monetization platform** (Google AdSense + direct-placement integration) generating ₹2L+/month in new digital revenue — including ad-serving logic, tracking, and reporting.
- **Implemented a responsive front-end redesign** that lifted mobile traffic share from 45% to 72%, matching the platform's shift to mobile-first consumption.

KEY PROJECTS

Generative AI Assistant Platform *(In Development)*

LLM-powered assistant automating content generation and business workflows — prompt pipelines, RAG-based context retrieval, and API-based model integration. Stack: React, Node.js, FastAPI, LangChain, ChromaDB, LLM APIs.

Clinic Appointment SaaS Platform *(Live)*

Multi-tenant appointment scheduling SaaS — online booking, doctor schedule management, OTP-based auth, role-based access, and automated notifications. Built with React, Node.js, and MySQL; hardened via a dedicated security audit and end-to-end test suite.

B2B E-commerce Platform — 5,000+ SKUs *(Live)*

Full-featured B2B commerce application — bulk ordering, tiered pricing, secure authentication, and Razorpay payment integration; optimized catalog search and listing for large product volumes with a responsive, mobile-first storefront.

EDUCATION

Master of Computer Applications (MCA)

2022

Siliguri Institute of Technology, Siliguri — focus on software engineering, web technologies, and cloud computing

CERTIFICATIONS

- Full Stack Development & Product Engineering — PESTO, 2024
- Cloud Computing — NPTEL (IIT Kharagpur), 2022
- AWS Certified Solutions Architect – Associate — Amazon Web Services (AWS), 2026
- Docker Foundations Professional Certificate – LinkedIn